

# Subhash Chandra

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## Education

**D.Y. Patil Deemed to be University** July 2023 – June 2027  
B.Tech in Computer Science and Engineering (Major: Artificial Intelligence & Machine Learning) Navi Mumbai, India  
**Relevant Coursework:** Machine Learning, Deep Learning, Data Structures & Algorithms, Database Management Systems, Statistics, Linear Algebra, Computer Vision, Natural Language Processing

## Experience

**Machine Learning Freelancer** – Outlier.ai 2024 – Present  
Remote

- Trained and fine-tuned 15+ machine learning models including classification, regression, and NLP tasks with average accuracy improvement of 12% through hyperparameter optimization.
- Performed comprehensive data preprocessing on datasets exceeding 100K+ records using Python (Pandas, NumPy), handling missing values, feature engineering, and data normalization.
- Evaluated model performance using cross-validation, confusion matrices, ROC-AUC scores, and precision-recall metrics, ensuring production-ready model quality.
- Achieved \$109 in earnings while maintaining 95% client satisfaction rate through timely delivery and high-quality ML solutions.
- Collaborated with international clients on diverse ML problems including time-series forecasting, image classification, and sentiment analysis.

## Projects

**Object Detection using R-CNN** | Python, TensorFlow, PyTorch, OpenCV, Flask November 2024  
*GitHub:* [github.com/subhash/object-detection](https://github.com/subhash/object-detection)

- Developed end-to-end object detection system using pre-trained Faster R-CNN (ResNet-50 backbone) achieving 87% mAP on COCO validation dataset with 5000+ images across 80 object categories.
- Implemented real-time image upload interface with Flask API processing 30+ images per minute with average inference time of 1.2 seconds per image.
- Utilized transfer learning and fine-tuning techniques reducing training time by 40% while maintaining detection accuracy above 85% on custom dataset.
- Applied advanced image preprocessing using OpenCV including normalization, augmentation (rotation, flipping, scaling), and annotation management improving model robustness by 15%.
- Deployed model with confidence threshold tuning (0.5-0.9) and Non-Maximum Suppression (NMS) for optimal bounding box predictions with IoU threshold of 0.4.

**Machine Learning Recommendation System** | Python, Scikit-learn, Pandas, NumPy, Flask September 2024  
*GitHub:* [github.com/subhash/recommendation-system](https://github.com/subhash/recommendation-system)

- Built hybrid recommendation system combining collaborative filtering (SVD algorithm) and content-based filtering serving 10K+ user-item interactions with 91% recommendation relevance score.
- Implemented matrix factorization using Singular Value Decomposition (SVD) reducing dimensionality from 5000 to 100 features while preserving 95% of variance.
- Engineered 25+ features including user demographics, item metadata, temporal patterns, and user behavior sequences improving prediction accuracy by 18%.
- Achieved Mean Absolute Error (MAE) of 0.68, Root Mean Squared Error (RMSE) of 0.82, and Precision@10 of 0.89 through rigorous hyperparameter tuning.
- Optimized model pipeline using parallel processing and caching strategies reducing recommendation generation

time from 8 seconds to 1.2 seconds (85% improvement).

- Deployed REST API using Flask handling 500+ concurrent requests with average response time under 200ms.

**Autonomous Snake Game using Deep Q-Learning** | Python, TensorFlow, OpenAI Gym, Pygame

July 2024

GitHub: [github.com/subhash/rl-snake](https://github.com/subhash/rl-snake)

- Created reinforcement learning agent using Deep Q-Network (DQN) with experience replay achieving 82% win rate and average score of 45 points after 1000+ training episodes.
- Designed custom reward function balancing food collection (+10), survival time (+1), and collision penalty (-100) resulting in 3x faster convergence compared to baseline implementation.
- Implemented epsilon-greedy exploration strategy (epsilon decay: 1.0 to 0.01 over 800 episodes) balancing exploration-exploitation tradeoff for optimal policy learning.
- Built neural network with 3 hidden layers (128-64-32 neurons) using ReLU activation processing 11-dimensional state space (distances, directions, dangers) predicting 4 action Q-values.
- Tuned hyperparameters including learning rate (0.001), discount factor (0.95), batch size (1000), and replay memory size (100K) improving average game score by 156% over 500 episodes.
- Visualized training metrics using Matplotlib tracking episode rewards, moving average scores, and epsilon decay demonstrating clear learning progression.

## Technical Skills & Certifications

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**Programming Languages:** Python (Advanced), Java (Intermediate), C (Intermediate), JavaScript, HTML5, CSS3, SQL

**Machine Learning & AI:** Supervised Learning (Classification, Regression), Unsupervised Learning (Clustering, Dimensionality Reduction), Reinforcement Learning (Q-Learning, DQN), Transfer Learning, Hyperparameter Tuning, Cross-Validation, Feature Engineering

**Deep Learning:** Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Transfer Learning, Model Fine-tuning, Computer Vision, Image Processing, Object Detection (R-CNN, YOLO), Image Classification

**Frameworks & Libraries:** TensorFlow, PyTorch, Keras, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, OpenCV, OpenAI Gym, NLTK

**Tools & Technologies:** Git, GitHub, Jupyter Notebook, Google Colab, VS Code, Flask, FastAPI, REST APIs, Docker (Basic)

**Databases & Query:** SQL, MySQL, PostgreSQL, Database Design, Query Optimization

**Data Science:** Data Preprocessing, Exploratory Data Analysis (EDA), Data Visualization, Statistical Analysis, A/B Testing, Feature Selection

**Core CS Fundamentals:** Data Structures, Algorithms, Object-Oriented Programming (OOP), Problem Solving, Time & Space Complexity Analysis, Design Patterns

**edX – Amazon Web Services:** Generative AI and Large Language Models on AWS

*Issued May 2025*

**Skills:** Generative AI, Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), AI Agent Building, Model Fine-tuning, AWS SageMaker, Model Optimization

**Certificate:** View Credential

**IBM via edX:** Introduction to Web Development – HTML5, CSS3, JavaScript

*Issued May 2025*

**Skills:** HTML5, CSS3, JavaScript, Responsive Web Design, Front-End Development, DOM Manipulation

**Certificate:** View Credential

**IBM via edX:** Python Basics for Data Science

*Issued April 2025*

**Skills:** Python Programming, Data Science Fundamentals, Pandas, NumPy, Data Manipulation, Data Analysis, Data Visualization

**Certificate:** View Credential

## Leadership & Extracurricular Activities

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**AI/ML Student Community – Active Member**

2024 – Present

D.Y. Patil University

- Actively participate in weekly AI/ML workshops, technical discussions, and peer learning sessions with 50+ community members.
- Contributed to knowledge-sharing initiatives by presenting projects on Deep Learning and Reinforcement Learning to 30+ students.

- Collaborate on group coding challenges and algorithm practice sessions improving problem-solving skills.

## Technical Blogger & Content Creator

2024

Personal Blog & LinkedIn

- Author technical blog posts on Machine Learning concepts, project tutorials, and AI trends reaching 200+ readers monthly.
- Create educational content simplifying complex ML algorithms (CNNs, RNNs, Q-Learning) with practical code examples.

## Achievements

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- **Outlier.ai Freelance Success:** Completed 15+ ML projects with 95% client satisfaction, earning \$109 through high-quality deliverables.
- **Project Portfolio:** Built 8+ end-to-end ML projects spanning Computer Vision, Recommendation Systems, and Reinforcement Learning.
- **Continuous Learning:** Completed 3 industry-recognized certifications in AI/ML and web development within 6 months.
- **Open Source Contribution:** Actively maintaining GitHub repositories with documented code and project demonstrations.

## Interests

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Competitive Programming, Machine Learning Research, Open Source Contributions, Tech Blogging, Problem Solving (LeetCode), Chess, Reading AI Research Papers